

```

    }
}
private:
    int val;
};

```

Here, we use 3 states instead of 2: 0 – unlocked, 1 – locked + no contention, and 2 – locked + contention. Also, this code uses `cmpxchg()`, which also returns the old value. If the lock was free, the thread simply gets the lock, sets it to 1 and returns. If the lock was already taken, there are two cases:

a) In the uncontended case, nobody else is contesting for the lock; hence, its value is 1. We indicate that there's a waiting thread by setting the value to 2.

b) In the contended case, the value is already 2. The check (`c==2`) succeeds and the thread simply blocks. But if the `futex_wait()` fails (because of `unlock()` code waking it up, or an interrupt), then it would have to try to set the value of lock again. But here (in the `while()` statement), we try to set it from 0 to 2 directly, since we don't know at this point if we are the only ones who are locking. There may be many waiting.

Now, in the lock code there's no question of overflow, since the highest value it reaches is 2. This fixes both correctness problems 1) and 2) discussed above. In the unlock code, check that the state is contended, and only then call `futex_wake()`. This fixes the performance problem discussed earlier.

Kernel support for futexes

At the kernel level, each futex has a unique ID, which is the *struct* page pointer and the offset within the page. This way, even though the futex variable may have different virtual addresses across processes, the physical address is the same, hence providing synchronisation. The reference count of the page is incremented, so that it is not swapped out of memory even when the process is sleeping. The futex structure on which a process is sleeping is placed on the kernel stack of the process.

A hash table maps the futex ID (`uaddr` param, above) to per-thread/per-process `futex_q` structs. When `futex_wait()` is called by a thread, a new `futex_q` object is created for it and added to the appropriate hash table entry. When `futex_wake()` is called, the same is used to wake up the sleeping process.

Do refer to the links below about futex implementation at the kernel level, for more details. 

References

- [0] <http://www.cs.temple.edu/~giorgio/cis307/readings/futex0.pdf>
- [1] <http://www.cs.temple.edu/~giorgio/cis307/readings/futex.pdf>
- [2] http://locklessinc.com/articles/mutex_cv_futex/
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